

# Differential Calculus

## \*Home work sheet # 9

### Directional Derivatives

**1–8** Find  $D_{\mathbf{u}}f$  at  $P$ . ■

1.  $f(x, y) = (1 + xy)^{3/2}$ ;  $P(3, 1)$ ;  $\mathbf{u} = \frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$

2.  $f(x, y) = \sin(5x - 3y)$ ;  $P(3, 5)$ ;  $\mathbf{u} = \frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}$

3.  $f(x, y) = \ln(1 + x^2 + y)$ ;  $P(0, 0)$ ;

$$\mathbf{u} = -\frac{1}{\sqrt{10}}\mathbf{i} - \frac{3}{\sqrt{10}}\mathbf{j}$$

4.  $f(x, y) = \frac{cx + dy}{x - y}$ ;  $P(3, 4)$ ;  $\mathbf{u} = \frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j}$

5.  $f(x, y, z) = 4x^5y^2z^3$ ;  $P(2, -1, 1)$ ;  $\mathbf{u} = \frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$

6.  $f(x, y, z) = ye^{xz} + z^2$ ;  $P(0, 2, 3)$ ;  $\mathbf{u} = \frac{2}{7}\mathbf{i} - \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}$

7.  $f(x, y, z) = \ln(x^2 + 2y^2 + 3z^2)$ ;  $P(-1, 2, 4)$ ;

$$\mathbf{u} = -\frac{3}{13}\mathbf{i} - \frac{4}{13}\mathbf{j} - \frac{12}{13}\mathbf{k}$$

8.  $f(x, y, z) = \sin xyz$ ;  $P\left(\frac{1}{2}, \frac{1}{3}, \pi\right)$ ;

$$\mathbf{u} = \frac{1}{\sqrt{3}}\mathbf{i} - \frac{1}{\sqrt{3}}\mathbf{j} + \frac{1}{\sqrt{3}}\mathbf{k}$$

**9–18** Find the directional derivative of  $f$  at  $P$  in the direction of  $\mathbf{a}$ . ■

9.  $f(x, y) = 4x^3y^2$ ;  $P(2, 1)$ ;  $\mathbf{a} = 4\mathbf{i} - 3\mathbf{j}$

10.  $f(x, y) = 9x^3 - 2y^3$ ;  $P(1, 0)$ ;  $\mathbf{a} = \mathbf{i} - \mathbf{j}$

11.  $f(x, y) = y^2 \ln x$ ;  $P(1, 4)$ ;  $\mathbf{a} = -3\mathbf{i} + 3\mathbf{j}$

12.  $f(x, y) = e^x \cos y$ ;  $P(0, \pi/4)$ ;  $\mathbf{a} = 5\mathbf{i} - 2\mathbf{j}$

13.  $f(x, y) = \tan^{-1}(y/x)$ ;  $P(-2, 2)$ ;  $\mathbf{a} = -\mathbf{i} - \mathbf{j}$

14.  $f(x, y) = xe^y - ye^x$ ;  $P(0, 0)$ ;  $\mathbf{a} = 5\mathbf{i} - 2\mathbf{j}$

15.  $f(x, y, z) = xy + z^2$ ;  $P(-3, 0, 4)$ ;  $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

16.  $f(x, y, z) = y - \sqrt{x^2 + z^2}$ ;  $P(-3, 1, 4)$ ;

$$\mathbf{a} = 2\mathbf{i} - 2\mathbf{j} - \mathbf{k}$$

17.  $f(x, y, z) = \frac{z - x}{z + y}$ ;  $P(1, 0, -3)$ ;  $\mathbf{a} = -6\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$

18.  $f(x, y, z) = e^{x+y+3z}$ ;  $P(-2, 2, -1)$ ;  $\mathbf{a} = 20\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$

\*These problems are for the students only as home work. Search the reference books for more examples.